# WAZP

MONOPHONIC SYNTHESIZER

## Owner's Manual

Version 1.0.4




TICCO ROSS | ticcoross.gumroad.com

# Welcome

WAZP is a monophonic synthesizer plug-in inspired by the small, buzzy British synths of the late 1970s: two raw oscillators, a screaming state-variable filter, a control oscillator for wobbles and random bleeps, and two envelopes that can repeat on their own.




## Formats

- macOS: Audio Unit, VST3 and standalone app (Apple Silicon and Intel), one .pkg installer.
- Windows: VST3 and standalone app, one Setup.exe installer.
- Linux: VST3, LV2 and standalone app (.tar.gz with an install script).

## Installing

**macOS:** open *WAZP-1.0.4-macOS.pkg* and follow the steps. Click **Customize** to choose formats.

> ⚠ **macOS security warning.** The WAZP installer is not notarised by Apple, so the first time you open it macOS may say it “cannot be opened because it is from an unidentified developer” or “cannot be verified”. This is expected. To install: **right-click** (or Control-click) the .pkg and choose **Open**, then click **Open** again. If macOS still blocks it, go to **System Settings → Privacy & Security**, scroll down and click **Open Anyway** next to the WAZP message.

**Windows:** run *WAZP-1.0.4-Windows-Setup.exe*. The VST3 goes to *C:\Program Files\Common Files\VST3*.

**Linux:** unpack the .tar.gz and run *./install.sh*.

Then rescan plug-ins in your DAW (Ableton Live: Settings → Plug-Ins → Rescan) and load WAZP on a MIDI track.

# The Panel

Drag a knob up or down to turn it (scroll wheel works too), double-click to return it to its default. While you move a control, the preset window shows its name and value. The window can be resized from the bottom-right corner.

## Keyboard

| Control | What it does |
|---|---|
| **SYNC** | Hard-syncs oscillator 2 to oscillator 1. Set oscillator 2 higher than oscillator 1 (range or PITCH) and sweep PITCH for the classic tearing sync sound. |
| **GLIDE** | Portamento: how long the pitch takes to slide to a new note. Fully left = off, up to about 2 seconds. |
| **Keys** | The on-panel keyboard is playable with the mouse (C3 - C5) and lights up with incoming MIDI notes. MIDI pitch bend (+/- 2 semitones) and the mod wheel (adds pitch modulation) also work. |

## Oscillators

| Control | What it does |
|---|---|
| **FT. (both)** | Footage / octave range: 32′, 16′, 8′, 4′, 2′. 8′ plays at the pitch of the key. |
| **WIDTH %** | Pulse width of oscillator 1, from 10 % to 90 % (50 % is a square wave). Only heard with the pulse waveform. |
| **Wave (both)** | OFF, pulse (⊓) or sawtooth. OFF silences the oscillator. |
| **PITCH** | Tunes oscillator 2 against oscillator 1, +/- 12 semitones. Small offsets give a fat detuned sound; 7 = a fifth, 12 = an octave. |

## Mixer / Output

| Control | What it does |
|---|---|
| **VOLUME** | Master output level, -80 dB to +6 dB. |
| **OSC 1, OSC 2** | Level of each oscillator going into the filter (0 - 10). Push both up for more filter drive and grit. |

# The Panel (continued)

## Control Osc.

| Control | What it does |
|---|---|
| **FREQ.** | Speed of the control oscillator, 0.05 Hz up to 1 kHz. It restarts with every new note. Above about 30 Hz it stops being a wobble and becomes audio-rate modulation: metallic, growling, FM-like tones. |
| **PITCH MOD.** | How much the control oscillator moves the pitch of both oscillators (up to +/- 12 semitones). Vibrato at low settings, sirens and bleeps higher up. |
| **NOISE SIGNAL** | Level of white noise in the mixer (0 - 10). Use it alone with the filter for wind, surf and percussion. |
| **Wave** | Shape of the control oscillator: sine, square, noise (smoothed random movement) and RND (a new random value every cycle: sample and hold). |

## Filter

| Control | What it does |
|---|---|
| **FREQ.** | Filter cutoff frequency, 20 Hz to 20 kHz. |
| **Q** | Resonance. The filter is two stages in series, so it rings hard and screams; at the top it self-oscillates and whistles by itself, even with the oscillators off. |
| **CONTROL** | Filter type: LO (low-pass), BAND (band-pass) or HI (high-pass). |
| **Left -/+ knob** | Amount of the control oscillator on the cutoff (wah, wobble, random filter steps), up to +/- 8 kHz. With a low FREQ. setting a deep amount can close the filter completely on the low part of the cycle, giving a gated, chopped sound. Centre = no modulation. |
| **Right -/+ knob** | Amount of the CONTROL ENV. on the cutoff. Right of centre opens the filter, left of centre closes it (inverted envelope). |

## Envelope Generators

| Control | What it does |
|---|---|
| **VCA ENV. ATTACK / DECAY** | How fast the sound rises when you press a key and how fast it falls to the sustain level. DECAY also sets the release time after you let go. |
| **REPEAT / SUSTAIN LEVEL** | Fully left (REPEAT) the envelope keeps re-triggering itself while a key is held: automatic repeating notes. Turned right, it sets the level the sound holds at while the key is down. |
| **CONTROL ENV. ATTACK / DECAY** | The envelope that sweeps the filter (through the left -/+ knob). It decays to zero. |
| **REPEAT / DELAY** | Fully left (REPEAT) the control envelope loops while the key is held (rhythmic filter sweeps). Turned right, it delays the start of the control envelope by up to 2 seconds. |

# Presets

The preset window at the top right shows the current preset number and name.

| Control | What it does |
|---|---|
| < > | Step through the factory presets, followed by your own presets. |
| **Preset display** | Click the name to open the full preset list: all factory presets, then your own presets (MY PRESETS). The current one is ticked; click any name to load it. OPEN PRESET FOLDER at the bottom shows your presets in Finder / Explorer. |
| **SAVE** | Saves the current sound as a .wazp file. By default it goes to *Documents / Ticco Ross / WAZP / Presets*; presets saved there appear in the < > list. |
| **LOAD** | Opens any .wazp file. |

WAZP comes with 16 factory presets: INIT, BUZZ BASS, WASP STING, SYNC SCREAM, REPEAT PERC, DREXY ACID, NOISE WIND, SIREN, RANDOM BLEEPS, HOLLOW LEAD, HIGH PASS FIZZ, BAND PLUCK, SUB DRONE, WOBBLE, SLOW SWELL and 8-BIT SQUARE. The presets also appear in your DAW's program list.

> Your DAW saves the complete state (every knob and the preset name) with the project, so you don't need to save a preset to keep a sound in a song.

# Automation & MIDI

All controls can be automated and MIDI-mapped from your DAW. WAZP responds to MIDI notes on any channel, pitch bend (+/- 2 semitones) and the mod wheel (CC 1, adds pitch modulation from the control oscillator). Velocity is ignored, as on the original instrument.

# Sound Design Tips

- **Classic buzz:** both oscillators on sawtooth, PITCH slightly off zero, Q around 50 %, a short CONTROL ENV. on the filter.
- **Sync lead:** SYNC on, oscillator 2 one range higher, then automate PITCH.
- **Self-playing rhythm:** VCA REPEAT fully left with a short DECAY gives repeating notes; add CONTROL ENV. REPEAT for moving filter pulses.
- **Bleeps:** control-osc wave on RND, FREQ. around 6 - 10 Hz, PITCH MOD. up.
- **Noise percussion:** oscillators OFF, NOISE SIGNAL up, BAND filter with high Q, short VCA decay.

# About & Support

Click the WAZP logo to open the About window, which shows the version, links to the store and contact address, and opens this manual.



Store: **ticcoross.gumroad.com** Contact: **ticcoross@gmail.com**